QuickCompany.in

Login

Keywords, Application#, Patent#, PCT#

< Back

An Improved Process For The Preparation Of Bland Turmeric Powder With Enhanced Curcuminoids

Abstract: Turmeric (Curcuma longa L.) belongs to the family Zingiberaceae, has been known for its flavouring/colouring properties. India has 149410 hectares area under cultivation with a annual production of 5,27,960 tones.. The commercially important turmeric products are turmeric powder and oleoresin. However, the above products have strong flavour and bitter taste, hence undesirable for colouring/flavouring food products. Thus, the oleoresin is further subjected to crystallization of relatively high purity colourant i.e. water insoluble curcuminoids (90-95 %) and generates waste called curcumin removed turmeric oleoresin (CRTO) having about 15-16% uncrystallisable curcuminoids. The present new colourant (bland turmeric powder) is prepared by treating turmeric powder with two selected solvents/solvent mixture in two stages to selectively remove non-colouring constituents (viz. bitter principles, aroma, fatty oil, resinous matter etc.).

QuickCompany.in

Login

bakery products, puddings, beverages etc., which over comes the problem of application of pure curcuminoids in aqueous preparation and also CRTO waste generation. The colourant containing 2.5% curcuminoids required to impart optimum yellow colour to above said preparations is in the range of 0.05 to 0.15%. The by-product is light yellowish mild turmeric flavoring, which can be used in food and pharmaceutical applications.

Get Free WhatsApp Updates!
Notices, Deadlines & Correspondence

FOLLOW

Patent Information

Application # 1689/DEL/2010	Filing Date 20 July 2010
Publication Number 42/2013	Publication Type INA
Invention Field CHEMICAL	Status Granted
Email	Parent Application
Patent Number 355145	Legal Status Ceased
Grant Date 2021-01-01	Renewal Date 20 Jul 2023

QuickCompany.in

Login

1. COUNCIL OF SCIENTIFIC & INDUSTRIAL RESEARCH
ANUSANDHAN BHAWAN, RAFI MARG, NEW DELHI - 110 001, INDIA.

Inventors

1. J. PURA NAIK
CFTRI, MYSORE-20

2. S.R. SAMPTHU
CFTRI, MYSORE-20

3. M. MADHAVA NAIDU
CFTRI, MYSORE-20

4. G. SULOCHANAMMA
CFTRI, MYSORE-20

Claims

An Improved Process For The P...
Council Of Scientific & Industrial Research
Info | Spec | Claims | Docs 31 | ERegister

700 microns; b) loading the powder as obtained in step [a] in an extraction column followed by extraction with a solvent mixture comprising of 94.98 to 74.50 parts of water in 5-25 parts of methanol and 00.02 to 00.50 parts of an organic/mineral acid followed by aq. methanol at temperature ranging from 0 to 60°C to obtain extract 1 and turmeric spent; c) drying the turmeric spent as obtained in step [b] at a temperature in the range of 50 to 60 degree C and loading to a fresh column followed by extracting with a hydrocarbon non-polar solvent to obtain an extract 2 and spent

QuickCompany.in Login

obtain bland turmeric powder with enhanced curcuminoids; e) simultaneously, neutralizing the extract 1 as obtained in step [b] with an alkali followed by desolventization to obtain a brownish resinous matter; f) optionally, desolventizing the extract 2 as obtained in step [c] to obtain a yellowish aromatic flavourant.

2. A process as claimed in claim 1, wherein the moisture level of the turmeric fingers is preferably brought down to 4 to 6%.

3. A process as claimed in claim 1, wherein drying is preferably carried out using a hot air drier.

4. A process as claimed in claim 1, wherein the organic and/or mineral acids are preferably selected from the group consisting of citric acid, acetic acid and hydrochloric acid.

5. A process as claimed in claim 1, wherein the hydrocarbon non-polar solvent is preferably selected from petroleum ether and hexane.

6. A process as claimed in claim 1, wherein the bland turmeric powder with enhanced curcuminoids is useful as a coloring agent in foods to impart a natural orange to yellow colour for fat soluble applications.

7. An improved process for the preparation of bland turmeric powder with enhanced curcuminoids substantially as herein described with reference to the foregoing examples.

Specification

Field of the Invention
The present invention relates to an improved process for the preparation of bland turmeric powder with enhanced curcuminoids from turmeric {Curcuma longa L).

QuickCompany.in Login

throughout the tropics but is predominantly Asian. The well-established, rhizomatous, aromatic economic spices of this family are the genus Zingier (ginger), Curcuma (turmeric), Alpinia (galangal) and Kaemfera. Turmeric plants belong to rhizomatous Curcuma species and have been known for their flavouring and colouring properties. Turmeric is grown mainly in India, China, Indonesia, Malaysia, Sri Lanka, Jamaica and Peru. Turmeric is one of the major spices grown in India. In India its annual production was 8,30,200 MT from an area of 1,74,600 hectares in 2005-06 and export was 51,500 MT valued Rs 16,480 lakhs in 2006-07.

After harvesting, the rhizomes are subjected to a process comprising cooking in water and drying in the sun to produce a product of fairly uniform colour. The cured turmeric fingers so obtained are hard, brittle and yellow in colour due to the presence of the phenolic compounds called curcuminoids. The curcuminoids, which are present in an oily cell phase, are mainly curcumin (reddish orange) with two methoxy group, demethoxycurcumin (orange-yellow) with one methoxy group and bis - demethoxycurcumin (yellow) without any methoxy group. The cured turmeric is marketed as bulbs and fingers for use as a spice and also as a bulk raw material for commercial processing.

The commercially important turmeric products are turmeric powder, oleoresin and curcuminoids. Oleoresin extracted from turmeric powder using solvents such as methyl ethyl ketone, acetone, ethylene dichloride, comprises essential oil, fatty oil and curcuminoids (30-50% by weight). However, the essential oil fraction of the turmeric/oleoresin has a very strong flavour and bitter taste and as such is not desirable for colouring food products. In order to meet the increasing demand for a flavour and taste free curcuminoid (neutraceutical) rich product, the oleoresin may be processed further. Thus, the oleoresin is subjected to crystallization step to obtain curcuminoid rich powder of a relatively high purity (90-95 % by weight) in respect of curcuminoid. The residual material that remains after the separation of curcuminoid crystals called mother liquor or curcumin removed turmeric oleoresin (CRTO) is a by product/industrial waste which contain about 15-16% curcuminoid. However, this residual material is unsuitable as a

QuickCompany.in Login

colourant commercially. It is noteworthy to mention here that curcuminoid (30-50ppm) is a promising substitute for the synthetic colour tartrazine (80-100ppm) in the food products.
There are several patents on turmeric colour and its applications. Reference may be made to Indian patent application No. 401/DEL/01 wherein provided is a process for the preparation of formulation using turmeric oleoresin for food application. Indian patent application No. 168/DEL/02 states the method of recovery of curcuminoid from industrial waste (CRTO) and IN194592 provides methods for pigment (curcuminoid) stabilization or prevention of fading.
Reference may be drawn to JP 2007037448 which gives a method for reducing bitter taste of turmeric extract. WO200704358 provides a process for producing enriched fractions of tetrahydroxy-curcumin from the extracts of Curcuma longa. US4163803 recites a process on eliminating the bitter principal from turmeric or extractives of turmeric such as solvent extracts or debittered solvent extracts by the addition to glycine. JP 200312578 provides a process to prepare turmeric alcohol extract to prepare refreshing beverages.
Turmeric is conventionally processed industrially by the solvent extraction process using solvents like acetone or ethylene dichloride to obtain a resinous mass called oleoresin. The conventional oleoresin has 25-40% curcumin, the rest being volatile oil, fatty oil and resinous constituents. Pure curcumin is recovered from the oleoresin by crystallization method using appropriate solvents to overall recovery of curcumin being 50-55% on raw material basis (turmeric powder). Thereof 45-50% of curcumin remains unrecovered in the mother liquor called CRTO (curcumin removed turmeric oleoresin). It is desirable to develop alternative processing approaches for utilizing of turmeric to produce colourants and flavourants.
Reference may further be made to US 6576273 and US 6942881, wherein the starting material used is curcumin removed turmeric oleoresin [CRTO] - an industrial waste containing non-crystallizable curcuminoids to prepare curcuminoids enriched coloring agent using hydrocarbon and pure methanol following phase separation technique. However, in the present invention, the curcuminoids are retained in the turmeric matrix by selectively removing the undesirable aroma and bitter

QuickCompany.in Login

material used in the present invention is turmeric and not CRTO. Thus, keeping in purview the drawbacks of the hitherto reported prior art, the inventors of the present invention realized that there exists a need to provide a process which helps in the retention of curcuminoids in the turmeric matrix without imparting any odour and bitter taste to it and thus having the potential of utilization of total curcuminoids present in the turmeric matrix as natural food colourant in various foods and beverages.

The present invention, thus recites the preparation of the turmeric powder, free from non-colouring constituents viz., bitter principles, essential oil, fixed oil, resinous matter etc. This novel curcuminoids rich turmeric powder can be exploited as a colourant as well as a starting material for the extraction of curcumin/curcumin rich conserve.

Objects of invention

The main object of the present invention is thus to provide an improved process for the preparation of bland turmeric powder with enhanced curcuminoids with negligible alteration in the curcuminoids composition.

Another object of the present invention is to provide a process wherein turmeric powder is directly extracted in two stages with two selected solvents / solvent mixture to obtain a product bland turmeric powder (colourant) with enhanced curcuminoids and a liquid by-

product mild turmeric flavourant. Still another object of the present invention is to provide a bland turmeric powder (colourant) with enhanced curcuminoids, which can be directly used to impart yellow colour to food products prepared in powder/ semi-liquid form for example gillebi, boondi, milk, milk products, puddings etc. which over comes the problem of water soluble application.

Summary of invention

The present invention provides a process for the preparation of bland turmeric powder (colourant) with enhanced curcuminoids from turmeric. The two inventive extraction steps involved in this process are: (1) as a first step, the colour masking pigments and non colouring substances viz., brown colouring pigments, resinous matter, carbohydrates, bitter constituents etc. are selectively extracted from the turmeric powder leaving behind

QuickCompany.in Login

turmeric spent obtained from first step. The product obtained is a bright yellow, bland turmeric powder (colourant) with enhanced curcuminoids. The oleoresin obtained from the aforesaid turmeric powder (colourant) is curcuminoids rich conserve in powder form which can further be processed to prepare curcumin. The by-product obtained is yellowish, mild turmeric flavourant which is a mixture of essential oil and fatty oil.

Accordingly, the present invention provides an improved process for the preparation of bland turmeric powder with enhanced curcuminoids wherein the steps comprising:

a) drying the commercially available turmeric fingers at 50 to 60°C, to bring down the moisture level to 3 to 8% followed by grinding into a coarse powder so as to pass through a sieve of 500 to 700 microns;

b) loading the powder as obtained in step [a] in an extraction column followed by extraction with a solvent mixture comprising a mixture of 94.98 to 74.50 parts of water in 5-25 parts of methanol and 00.02 to 00.50 parts of an organic/mineral acid at temperature ranging from 0 to 60°C to obtain extract 1 and turmeric spent;

c) drying the turmeric spent as obtained in step [b] at a temperature in the range of 50 to 60 degree C and loading to a fresh column followed by extracting with a hydrocarbon non-polar solvent to obtain an extract 2 and spent material;

d) recovering the solvent from the spent material as obtained in step [c] followed by drying at a temperature in the range of 50 to 60 degree C to obtain bland turmeric powder with enhanced curcuminoids;

e) simultaneously, neutralizing the extract 1 as obtained in step [b] with an alkali followed by desolventization to obtain a brownish resinous matter;

f) optionally, desolventizing the extract 2 as obtained in step [c] to obtain a yellowish aromatic flavourant.

Detailed Description of the invention

The present invention provides a process for the preparation of bland turmeric powder (colourant) with enhanced curcuminoids from turmeric (Curcuma longa L), which results in the preparation of a commercially valuable turmeric powder (colourant) with enhanced colour value.

QuickCompany.in Login

demethoxy curcumin. Further, the turmeric powder obtained contains at least 0.50 to 3.86% curcuminoids more on raw material basis when compared to corresponding turmeric. However, the curcumin enrichment of the bland turmeric powder (colourant) depends on the variety of turmeric used. Also the curcuminoid content of the brownish resinous matter/non-colouring substances and byproduct a flavourant is almost negligible. The curcuminoids content is measured by the method prescribed by the Association of Officials American Chemists (AOAC).

The process of the present invention in the first step requires that a bland turmeric powder (colourant) with enhanced curcuminoids is provided. As used herein, the expression "bland turmeric powder (colourant)" includes material derived from plant species including plants of the genus Curcuma longa L, C. aromatica Salisb, C. amada Roxb., C. zedoaria Rose, and Xanthorrhiza Roxb. A curcuminoid containing starting material can be selected that has one of the above mentioned species / mixture of the species for example Curcuma longa L. and includes varieties like aleppy, Salem, Mysore local, North eastern region Lakadang etc. The variety Curcuma longa L used for the purposes of the present invention was procured from local market, Mysore, origin of Mysore District, Karnataka State (Sample - 1) and, Local Market, Lakadong, origin of North east region, India (Sample - 2).

Commercially available turmeric fingers of Curcuma longa (sample-1 and sample -2) were coarsely powdered in mechanical grinder. Further, this course powder was powdered in a hammer mill to 500 microns particles sizes separately were used as the raw material for extraction. The powder (100g) was loaded in a glass column and added 200 ml solvent mixture containing 0.02 parts of acid and 99.98 parts of 25% aqueous methanol by volume. The extract was eluted drop wise at room temperature (27°C) while simultaneously adding 500 ml fresh solvent mixture followed by adding 200ml of aqueous methanol in the column. The material in the column was removed and dried in a drier at 50-60°C. The dried material was loaded again in a fresh column and was extracted with 750 ml petroleum ether followed by collecting the extract drop wise. The column was connected to vacuum to recover absorbed petroleum ether from

QuickCompany.in Login

desolventized separately in a rotary evaporator at atmospheric pressure till nearly 95% of the solvent was recovered. Rest of the distillation was carried out under reduced pressure of 20" -26 inches and a temperature of less than 60°C using a suction pump till solvent was almost completely removed to obtain a yellowish aromatic flavourant (6.0 g). Similarly the acidic aqueous methanol extract was neutralized using alkali and desolventized to obtain brownish resinous matter (13.0g), which had negligible colour.

In an embodiment of the invention, draining of the extract and further addition of solvent mixture (acidic aqueous methanol) to selectively extract the non colouring substances, colour masking pigments, bitter constituents from the turmeric powder and elution of column is done using a material to solvent ratio of 1:5 to 1:10, till the eluant is almost colourless.

In another embodiment of the invention, the solvent mixture can be replaced with acidic water/ water.

In yet another embodiment of the invention, the solvents used are straight chain alcohols containing C1-C3 carbon atoms.

In still another embodiment of the invention, the acids used are mineral or organic acids.

In yet another embodiment of the invention, the essential oils and fatty oils are selectively extracted from the turmeric powder by draining of the extract and further addition of the hydrocarbon solvent to the column, and elution using a material to solvent ratio of 1:5 to 1:10, till the elute is almost free from turmeric flavour.

In still another embodiment of the invention, the essential oils and fatty oils are selectively extracted from the turmeric powder using hydrocarbon solvent viz. hexane, petroleum ether and the like.

In yet another embodiment of the invention, the product obtained is a bland turmeric powder (colourant) with enhanced curcuminoids having the following characteristics - (1) Appearance:
bright/dark yellow (2) Texture: free flowing powder (3) Aroma: negligible (4) taste: bland (5) Residual solvent: below 50 ppm. The parameters viz. curcumin content, oleoresin yield, ether extract, total soluble solids (TSS) are governed by the turmeric variety used.

QuickCompany.in Login

Specific gravity: 0.93401 (3) Optical rotation: + 3.132 (4) Refractive index: 1.5102 (5) Aroma: Mild turmeric (6) Residual solvent: below 50 ppm (7) Volatile oil content (% v/v): 60-70.

In another embodiment of the invention, the bland turmeric powder (colourant) with enhanced curcuminoids is subjected to desolventisation by indirect heating to remove and recover more than 95% of the absorbed solvent and subsequently dried.

In yet another embodiment of the invention, the extraction is carried out using stirrer, wherein the turmeric powder and solvent mixture are stirred in a suitable vessel and subjected the mass in to filtration or centrifugation.

In still another embodiment of the invention, the bland turmeric powder (colourant) with enhanced curcuminoids is suitable for use as a colouring agent in foods to impart a natural orange yellow to yellow colour for fat soluble applications.

In yet another embodiment of the invention, the colour constituents of the bland turmeric powder (colourant) with enhanced curcuminoids/curcuminoids conserve prepared as described is made water soluble by known art such as by treating with permitted emulsifiers viz. propylene glycol etc. which makes them useful for water soluble food applications.

In a further embodiment of the invention, the bland turmeric powder (colourant) with enhanced curcuminoids prepared as described is useful for non food applications such as cosmetics, pharmaceuticals and animal feeds.

In another embodiment of the invention, the byproduct liquid turmeric flavour prepared have mild turmeric aroma which is suitable to use as a turmeric flavouring agent in foods to impart desired level of turmeric aroma, but without any appreciable colour.

In yet another embodiment of the invention, the liquid turmeric flavourant is an essential oil and fatty oil mixture, which can be converted to powder form using, permitted diluents such as starch, gum arabica etc. to obtain flavouring formulations of required concentrations.

In still another embodiment of the invention, the bland turmeric powder (colourant) with enhanced curcuminoids is suitable for use as a raw material to prepare oleoresin in powder form (colourant) using permitted solvents.

In yet another embodiment of the invention, the oleoresin

QuickCompany.in Login

for use as a raw material to prepare pure curcumin adapting techniques such as crystallization, phase separation method etc. using suitable solvents.

EXAMPLES

The following examples are given by way of illustration and therefore should not be construed to limit the scope of the present invention.

Example 1

Commercially available turmeric fingers of Curcuma longa Mysore local (sample -1) were coarsely powdered in mechanical grinder. Further, this course powder was powdered in a hammer mill to 500 microns particles size was used as the raw material for extraction. The turmeric powder (lOOg) was loaded in a glass col umn and added 200 ml solvent mixture containing 0.02 parts of acid and 99.98 parts of 25% aqueous methanol by volume. The extract (650ml) was eluted drop wise at room temperature (27°C) while simultaneously adding 350ml fresh solvent mixture followed by 200ml of above solvent mixture except acid in the column. The material in the column was removed and dried in a drier at 50°C. The dried material was loaded again in a fresh column and was extracted with 750 ml petroleum ether followed by collecting the extract drop wise. The column was connected to vacuum to recover absorbed petroleum ether from the spent. The spent removed from the column was dried in drier at 50°C to obtain turmeric powder (colourant) with enhanced curcuminoids (80.0g). The hydrocarbon extract was desolventized separately in a rotary evaporator at atmospheric pressure till nearly 95% of the solvent was recovered. Rest of the distillation was carried out under reduced pressure of 20" - 26 inches and a temperature of less than 60°C using a suction pump till solvent was almost completely removed to obtain a yellowish aromatic flavourant (7.0 g). Similarly the acidic aqueous methanol extract was neutralized using alkali and desolventised to obtain brownish resinous matter (13.0g), which had negligible colour.

Example 2

The powder (sample - I) of 500 microns particles size 100g was loaded in a glass column and added 200 ml solvent mixture containing 0.2 parts of acid and 99.8 parts of 20% aqueous methanol by volume. The extract (650ml) was eluted drop wise at room temperature (27°) while simultaneously adding fresh

QuickCompany.in Login

60 °C. The dried material was loaded again in to the column was extracted with 750 ml petroleum ether collecting 650 ml extract drop wise. The column was connected to vacuum to recover absorbed petroleum ether from the spent. The spent removed from the column was dried in drier at 60 °C to obtain turmeric powder with enhanced curcuminoid (76.0g). The hydrocarbon extract was desolventized separately in a rotary evaporator at atmospheric pressure till nearly 95% of the solvent was recovered. Rest of the distillation was carried out under reduced pressure of 20" - 26 inches and a temperature of less than 60°C using a suction pump till solvent was almost completely removed to obtain a yellowish turmeric flavourant (4.6 g).

Similarly, the acidic aqueous methanol extract was neutralized using alkali and desolventised to obtain brownish resinous matter (18.8g), which had negligible colour

Example 3

Commercially available turmeric fingers of Curcuma longa Mysore local) were coarsely powdered in mechanical grinder. Further, this course powder was powdered in a hammer mill to was used as the raw material for extraction. The turmeric powder (sample- 1) of 500 microns particles size 100g was loaded in a glass column and added 200 ml solvent mixture containing 0.5 parts of acid and 99.5 parts of 33.33% aqueous methanol by volume. The extract was eluted drop wise at room temperature (27°C) while simultaneously adding 500 ml fresh solvent mixture followed by adding 200ml of above solvent mixture except acid in the column. The material in the column was removed and dried in a drier at 40°C. The dried material was loaded again in column and was extracted with 750 ml petroleum ether followed by collecting the extract drop wise. The column was connected to vacuum to recover absorbed petroleum ether from the spent. The spent removed from the column was dried in drier at 40 °C to obtain turmeric powder (colourant) (76.0g) with enhanced curcuminoids when compare to corresponding turmeric. The hydrocarbon extract was desolventized separately in a rotary evaporator at atmospheric pressure till nearly 95% of the solvent was recovered. Rest of the distillation was carried out under reduced pressure of 20" - 26 inches and a temperature of less than 60°C using a suction pump till solvent was almost completely removed to obtain a

QuickCompany.in Login

had negligible curcumin.

Example -4

The turmeric (sample - II) powder of 500 microns particles size 100 g was loaded in a glass column and added 200 ml solvent mixture containing 0.02 parts of acid and 99.98 parts of 25% aqueous methanol by volume. The extract (650ml) was eluted drop wise at room temperature (27°) while simultaneously adding fresh solvent mixture (350ml) followed by 200ml above solvent mixture except acid on to the top of the material in the column. The material in the column was removed and dried in a drier at 60°C. The dried material was loaded again in to the column was extracted with 750 ml petroleum ether followed by collecting 650 ml extract drop wise. The column was connected to vacuum to recover absorbed petroleum ether from the spent. The spent removed from the column dried in drier at 60 °C to obtain turmeric powder(76.g) with enhanced curcuminoid when compared to corresponding turmeric. The hydrocarbon extract was desolventized separately in a rotary evaporator at atmospheric pressure till nearly 95% of the solvent was recovered. Rest of the distillation was carried out under reduced pressure of 20" - 26 inches and a temperature of less than 60°C using a suction pump till solvent was almost completely removed to obtain a yellowish turmeric flavourant (7.4g).

Similarly, the acidic aqueous methanol extract was neutralized using alkali and desolventised to obtain brownish resinous matter (18.8g), which had negligible curcumin

Example 5

Commercially available turmeric fingers Lakadang variety was coarsely powdered in mechanical grinder. Further this coarse powder was powdered in a hammer mill to was used as the raw material for extraction. The turmeric (sample - II) powder of 500 microns particles size 140 g was loaded in a glass column and added 300 ml solvent mixture containing 0.4 parts of acid and 99.6 parts of 20% aqueous methanol by volume. The extract (950ml) was eluted drop wise at room temperature (27°) while simultaneously adding fresh solvent mixture (650ml) followed by 200ml above solvent mixture except acid on to the top of the

QuickCompany.in Login

again in to the column was extracted with 750 ml petroleum ether collecting 650 ml extract
drop wise. The column was connected to vacuum to recover absorbed petroleum ether from
the spent. The spent removed from the column dried in drier at 40°C to obtain turmeric powder
(108.8g) with enhanced curcuminoid (14.5%). The hydrocarbon extract was desolventized
separately in a rotary evaporator at atmospheric pressure till nearly 95% of the solvent was
recovered. Rest of the distillation was carried out under reduced pressure of 20" - 26 inches
and a temperature of less than 60°C using a suction pump till solvent was almost completely
removed to obtain a yellowish turmeric flavourant (10.36 g). Similarly, the acidic aqueous
methanol extract was neutralized using alkali and desolventised to obtain brownish resinous
matter (119.5g), which had negligible colour The physico-chemical analysis of the products
(colourants) obtained from the above fiver experiments (examples 1-5) and the corresponding
turmeric samples i.e. I & II were carried out and the results are presented in table -1 & 2.

Table 1. Physico-chemical analysis data of "bland turmeric powder colourant" with enhanced curcuminoids
(Table Removed)

Table 2. Approximate composition*, residual solvent** content of turmeric powder colourant and corresponding turmeric powder
(Table Removed)
*Official Methods of analysis. A.O.A. C. Washington, 14th ed. (1984). **IS:5832-1975, ISI, Manak Bhavan, 9- Bhahadur Shah Zafar Marg, New Delhi-110002
Specification of the product/byproduct Product: Turmeric powder (colourant)
1. Nature: Curcuminoid rich colourant
2. Appearance: Bright yellow to dark yellow powder
3. Texture: Free flowing powder
4. Total soluble solid (%): 1.10-1.30
5. Moisture (%): 5.00-6.00

QuickCompany.in Login

9. Total sugars (%) 56.9 + 2
10. Protein (N X 6.25) (%) 8.7 + 1
11. Starch (%) 51.2 + 2
12. Crude fibre (%) 5.8, + 2
13. Total ash (%) 2.8 + 1
14. Petoleum ether (ppm) <25
15. Methanol (ppm) <50.

The parameters 6-13 varies with the variety of turmeric used.

FDA-US specification residual solvent limits for Methanol -<50 and Hexane-<25

By-product: Turmeric flavourant

(1) Nature : Rich in volatile & fatty oil constituents
(2) Appearance Yellowish free flowing liquid
(3) Specific gravity 0.93401
(4) Optical rotation +3.132
(6) Refractive index: 1.5102
(8)Aroma Mild turmeric aroma
(9) Residual solvent < 50ppm for Methanol &<25 ppm for(hexane)
(10) Volatile oil (%) : 60.00 to 70.00

Example-6

The varied applications of the turmeric powder colourant is described in the following examples

Preparation of turmeric powder (colourant): Turmeric powder colourant (2.3%curcuminoids) was uniformly ground in Brauwn dry grinder to 100 Mesh (BS) partical size for food application.

The above turmeric powder (colourant) was incorporated in food products/beverages to obtain
desired effects of low, medium, high colour or very high.

1). Preparation of sweet milk: 0.025g, 0.05g, 0.15g and 0.2g of turmeric powder colourant was mixed with 100ml portions of boiling milk containing lOg sugar for 5 minutes separately and the resultant were tested for optimum colour and sensory acceptance. The colour ratings of these milk beverages were: a). 0.025g-low b). 0.05g- medium c) O.lg-optimum d) 0.15g - high and e) 0.2g - very high. The turmeric aroma was not detected in all the above milk beverages prepared.

2). Preparation of sweet badam (almond) milk: 0.025g, 0.05g, O.lg, 0.15g and 0.2g of turmeric powder colourant was mixed with 100ml portions of boiling milk containing lOg sugar 2.0g

QuickCompany.in Login

medium c). 0.1g - optimum, d) 0.15g - high and e) 0.2g - very high. The turmeric aroma was not detected in all the above milk beverages prepared.

3). Preparation of kesari bath: 0.025, 0.05, 0.15 and 0.2% of turmeric powder colourant (2.5% curcumin), was used to prepare kesari bath using rava, milk, cashew, ghee separately and the resultant 4 kesari bath samples were tested for optimum colour and sensory acceptance. The colour ratings of these were: a). 0.025g -low, b). 0.05g - acceptable, c) 0.15g - optimum and d) 0.2g - high. The turmeric aroma was not detected in all the above kesari bath samples prepared.

4). Preparation of sweet boondi: 0.025, 0.05, 0.15 and 0.2% of turmeric powder colourant, was mixed with 150ml portions of boiling water for 5 minutes separately, cooled and prepared slurry with 100g Bengal gram dhal powder separately. These different slurries

prepared are used to prepare boondi separately. The resultant boondi samples were tested for optimum colour and sensory acceptance. The colour ratings of these boondi samples were: a). 0.025g - very low b). 0.05g - low , c) 0.15g - optimum and d) 0.2g - high. The turmeric aroma was not detected in all the above boondi samples prepared.

5). Preparation of pudding : 0.025, 0.05, 0.15 and 0.2% of turmeric powder colourant was mixed with pudding dry mix separately. 16g each pudding mix was mixed separately with 100ml portions of boiling milk and continued boiling for 5 minutes and transferred into 50 ml cups separately. The resultant puddings were tested for optimum colour and sensory acceptance. The colour ratings of these milk beverages were: a). 0.025g - low b). 0.05g -medium, c) 0.15g - high and d) 0.2g - very high. The turmeric aroma was not detected in all the above pudding samples tested.

Advantages of the invention:

• The bland turmeric powder (colourant) with enhanced curcuminoids can find varied applications as natural yellow colourant, neutraceutical etc in food / pharmaceutical industries.

• It is a facile process wherein turmeric powder is directly extracted in two stages with two selected solvents / solvent mixture to obtain a product bland turmeric powder (colourant) with enhanced curcuminoids and a liquid by-product mild

QuickCompany.in Login

colour conserve in powder form, since it is free from brown colouring pigments, non colouring substances bitter principles, flavour constituents, fatty oil, etc.

• The bland turmeric powder (colourant) with enhanced curcuminoids can be directly used to impart yellow colour to food products prepared in powder/ semiliquid form for example gillebi, boondi, milk, milk products, puddings etc. which over comes the problem of water soluble application.

• The byproduct obtained is having mild turmeric odor and aroma, which can be used to impart turmeric flavour of predetermined levels to food preparations. Also when used at normal or high aromatic levels it does not impart any appreciable colour or undesirable flavour to food products.

• The byproduct can also be used to prepare powder/liquid formulations of various flavour strengths for food application. In pharmaceutical preparations it can be used as antiseptic in ointments and bandages etc.

• The process is amenable for scale -up.

We claim:

1. An improved process for the preparation of bland turmeric powder with enhanced
curcuminoids wherein the steps comprising:

a) drying the commercially available turmeric fingers at 50 to 60°C, to bring down the moisture level to 3 to 8% followed by grinding into a coarse powder so as to pass through a sieve of 500 to 700 microns;

b) loading the powder as obtained in step [a] in an extraction column followed by extraction with a solvent mixture comprising of 94.98 to 74.50 parts of water in 5-25 parts of methanol and 00.02 to 00.50 parts of an organic/mineral acid followed by aq. methanol at temperature ranging from 0 to 60°C to obtain extract 1 and turmeric spent;

c) drying the turmeric spent as obtained in step [b] at a temperature in the range of 50 to 60 degree C and loading to a fresh column followed by extracting with a hydrocarbon non-polar solvent to obtain an extract 2 and spent material;

d) recovering the solvent from the spent material as obtained in step [c] followed by drying at a temperature in the range of 50 to 60 degree C to obtain bland turmeric powder with enhanced curcuminoids;

QuickCompany.in Login

f) optionally, desolventizing the extract 2 as obtained in step [e] to obtain a yellowish aromatic flavourant.
2. A process as claimed in claim 1, wherein the moisture level of the turmeric fingers is
preferably brought down to 4 to 6%.
3. A process as claimed in claim 1, wherein drying is preferably carried out using a hot air
drier.
4. A process as claimed in claim 1, wherein the organic and/or mineral acids are preferably
selected from the group consisting of citric acid, acetic acid and hydrochloric acid.
5. A process as claimed in claim 1, wherein the hydrocarbon non-polar solvent is
preferably selected from petroleum ether and hexane.
6. A process as claimed in claim 1, wherein the bland turmeric powder with enhanced
curcuminoids is useful as a coloring agent in foods to impart a natural orange to yellow
colour for fat soluble applications.
7. An improved process for the preparation of bland turmeric powder with enhanced
curcuminoids substantially as herein described with reference to the foregoing
examples.

Documents

ORDERS

Section	Controller	Decision Date
15 and 43	MEENAKSHI YADAV	2020-01-16
15 & 43	MEENAKSHI YADAV	2021-01-01

APPLICATION DOCUMENTS

QuickCompany.in

Login

2	1689-del-2010-form-3.pdf	2011-08-21
3	1689-del-2010-form-2.pdf	2011-08-21
4	1689-del-2010-form-1.pdf	2011-08-21
5	1689-del-2010-drawings.pdf	2011-08-21
6	1689-del-2010-description (complete).pdf	2011-08-21
7	1689-del-2010-correspondence-others.pdf	2011-08-21
8	1689-del-2010-claims.pdf	2011-08-21
9	1689-del-2010-abstract.pdf	2011-08-21
10	1689-del-2010-Form-18-(27-11-2012).pdf	2012-11-27
11	1689-del-2010-Correspondence Others-(27-11-2012).pdf	2012-11-27
12	1689-DEL-2010-FER.pdf	2017-08-04
13	1689-DEL-2010-OTHERS [29-01-2018(online)].pdf	2018-01-29
14	1689-DEL-2010-FER_SER_REPLY [29-01-2018(online)].pdf	2018-01-29
15	1689-DEL-2010-CLAIMS [29-01-2018(online)].pdf	2018-01-29
16	1689-DEL-2010-HearingNoticeLetter05-07-2019.pdf	2019-07-05
17	1689-DEL-2010-HearingNoticeLetter09-08-2019.pdf	2019-08-09
18	1689-DEL-2010-HearingNoticeLetter02-09-2019.pdf	2019-09-02

QuickCompany.in Login

20	1689-DEL-2010-Annexure (Optional) [03-09-2019(online)].pdf	2019-09-03
21	1689-DEL-2010-OTHERS-091219.pdf	2019-12-10
22	1689-DEL-2010-OTHERS-091219-.pdf	2019-12-10
23	1689-DEL-2010-Other Patent Document-091219.pdf	2019-12-10
24	1689-DEL-2010-Form 13-091219.pdf	2019-12-10
25	1689-DEL-2010-Claims-091219.pdf	2019-12-10
26	1689-DEL-2010-NBA Approval Submission [02-11-2020(online)].pdf	2020-11-02
27	1689-DEL-2010-PatentCertificate01-01-2021.pdf	2021-01-01
28	1689-DEL-2010-IntimationOfGrant01-01-2021.pdf	2021-01-01

SEARCH STRATEGY

1	searchstrategy_02-08-2017.pdf

ERegister / Renewals Ceased

3RD: 01 APR 2021

CBR 11640 | Renewal 01/04/2021 | Renewal Amount ₹4,000 | Certificate #22066

From 20/07/2012 - To 20/07/2013

4TH: 01 APR 2021

QuickCompany.in Login

From 20/07/2013 - To 20/07/2014

5TH: 01 APR 2021

CBR 11640 | Renewal 01/04/2021 | Renewal Amount ₹4,000 | Certificate #22068

From 20/07/2014 - To 20/07/2015

6TH: 01 APR 2021

CBR 11640 | Renewal 01/04/2021 | Renewal Amount ₹4,000 | Certificate #22069

From 20/07/2015 - To 20/07/2016

7TH: 01 APR 2021

CBR 11640 | Renewal 01/04/2021 | Renewal Amount ₹12,000 | Certificate #22070

From 20/07/2016 - To 20/07/2017

8TH: 01 APR 2021

CBR 11640 | Renewal 01/04/2021 | Renewal Amount ₹12,000 | Certificate #22071

From 20/07/2017 - To 20/07/2018

9TH: 01 APR 2021

CBR 11640 | Renewal 01/04/2021 | Renewal Amount ₹12,000 | Certificate #22072

From 20/07/2018 - To 20/07/2019

10TH: 01 APR 2021

QuickCompany.in

Login

From 20/07/2019 - To 20/07/2020

11TH: 01 APR 2021

CBR 11640 | Renewal 01/04/2021 | Renewal Amount ₹24,000 | Certificate #22074

From 20/07/2020 - To 20/07/2021

12TH: 14 JUL 2021

CBR 24613 | Renewal 14/07/2021 | Renewal Amount ₹24,000 | Certificate #48757

From 20/07/2021 - To 20/07/2022

13TH: 08 JUN 2022

CBR 22182 | Renewal 08/06/2022 | Renewal Amount ₹24,000 | Certificate #43295

From 20/07/2022 - To 20/07/2023

Home Trademark Search Company Search
Terms / Conditions Privacy About Contact Us